

Hawkeye EQMS · Internal Audit (8-phase lifecycle) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Contents

- 1. Module overview
- 2. Standard pharma EQMS vs Hawkeye — gap analysis
- 3. Personas + role mapping
- 4. Feature catalogue (click-by-click)
- 5. Full lifecycle walkthrough — one transaction, end-to-end
- 6. AI assistance map
- 7. Regulator-trace matrix
- 8. Test results matrix
- 9. Known gaps + roadmap

1. Module overview

Module	Internal Audit (8-phase lifecycle)
Pharma purpose	Run a supplier or internal audit through 8 sequential phases — INITIATED → PREP → PLANNING → EXECUTION → FINDINGS → CAPA → CLOSURE → SURVEILLANCE. Each phase owns specific artifacts. Role gates enforce buyer/supplier/auditor collaboration.
Compliance basis	ISO 19011 (auditing management systems) · ISO 9001:2015 §9.2 · 21 CFR 211.180 · ICH Q10 §3.2.4 · EU GMP Chapter 9
Hawkeye routes	/audits, /buyer/audits, /auditor/audits (frontend) · /api/audit-requests, /api/audits (backend)
Hawkeye model file	backend/src/models/auditRequestsMasterModel.js
Tenant module flag	modules.AUDIT_MANAGEMENT

An audit transitions through 8 phases sequentially. Each phase has phaseStatuses NOT_STARTED → IN_PROGRESS → COMPLETED → BLOCKED. Buyer creates the request, supplier accepts and runs PREP, auditor PLANS + EXECUTES + LOGS findings, supplier validates findings + submits CAPA, buyer reviews + signs CLOSURE certificate. AI assists at PREP (questionnaire generation), EXECUTION (auditor coach), and FINDINGS (report assembly with SHA-256 integrity hash).

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET
= parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
8-phase lifecycle (INITIATED → SURVEILLANCE) ISO 19011 §6	auditPhases.js with 8 enum values + sequential transition rule (canTransition).	MET	
Per-phase artifact tracking (intimation letter, DRL, agenda, checklist, PDR, CAPA, closure cert) ISO 19011	auditArtifactModel + auditArtifactVersionModel. Per-phase artifact list defined in auditPhaseService.	MET	
Buyer/supplier/auditor role gates ISO 19011	supplierDecision (PENDING/ACCEPTED/REJECTED/PROPOSED), auditorDecision, deficiencyValidation enums; role-based endpoints.	MET	
ISO 19011 finding severity (Critical / Major / Minor / Observation) ISO 19011 §6.4.8	GMP classification on auditReportModel observations.	MET	
AI questionnaire generation from supplier risk + past findings FDA AI guidance Jan 2025	POST /api/ai/audit-agents/prepare-questionnaire — risk-weighted from openFDA + past findings.	MET	
AI auditor-coach (real-time observation severity guidance) —	Wave 3 AuditorCoachPanel.	MET	
AI audit-report assembly with SHA-256 integrity hash 21 CFR Part 11 §11.10(c)	POST /api/ai/audit-agents/assemble-report. Output includes SHA-256 hash for tamper detection.	MET	
Closure certificate signed by buyer + supplier + auditor + witness (4 signatures) ISO 19011 §6.5.4	auditReportModel.signatures[] with 4 expected roles.	MET	
Surveillance follow-up scheduling ISO 19011 §6.6	SURVEILLANCE phase enables follow-up audit scheduling.	MET	
Conflict-of-interest declaration per auditor ISO 19011 §5.4.2	COI artifact in PLANNING phase.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Audit Program Mgr Priya buyer	<code>audit.program@novex-pharma.demo</code>	Initiates audit request, assigns auditors, awards RFQ, reviews findings, closes audit.	INITIATED FINDINGS CAPA CLOSURE
Lead Auditor Maria auditor	<code>audit.lead@novex-pharma.demo</code>	Plans audit, conducts execution, logs findings + observations, drafts report, signs closure.	PLANNING EXECUTION FINDINGS CAPA

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Supplier (Production Head Michael in demo) supplierUser / production	production.head@novex-pharma.demo	Accepts audit, runs PREP (DRL + SMF), validates / disputes findings, submits CAPA evidence.	PREP FINDINGS validation CAPA
Tenant Admin Elena tenant_admin · admin override	vp.quality@novex-pharma.demo	Force-transition phases when blocked; final witness signature on closure cert.	CLOSURE

4. Feature catalogue (click-by-click)

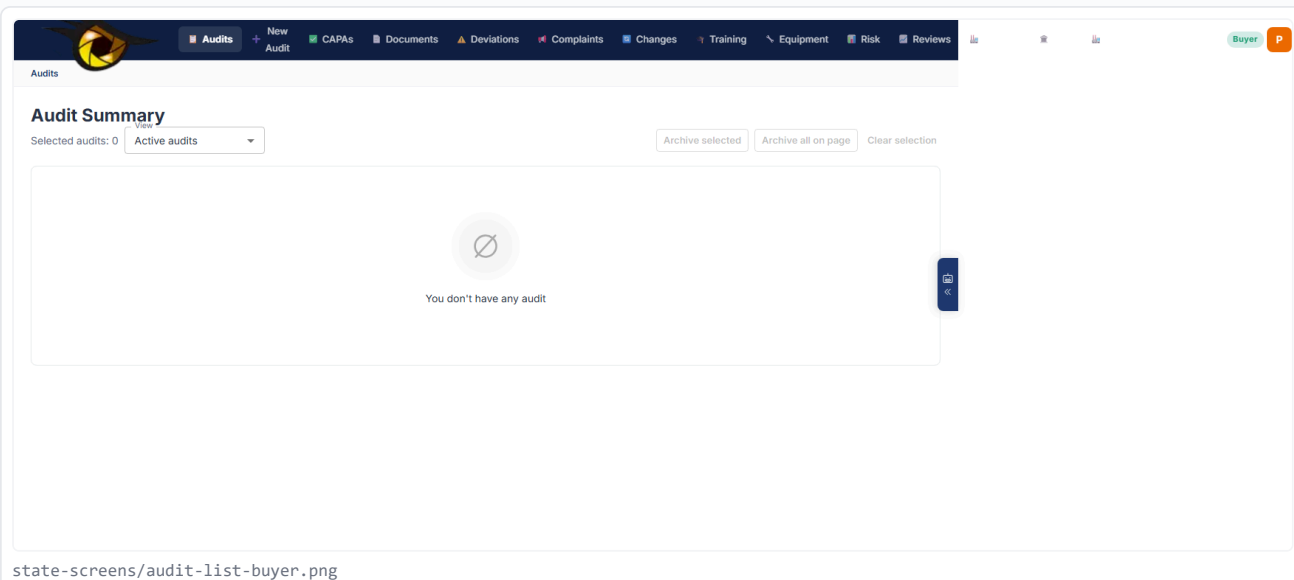
4.1 Audit register (buyer / auditor / supplier views)

What it does	List of audits scoped by role.
Menu / route	/buyer/audits (buyer) · /auditor/audits (auditor) · /audits (admin)
Available to roles	buyer auditor supplier admin
Backend API	GET /api/audit-requests/buyer · /auditor · /supplier

CLICK-BY-CLICK

- NAVIGATE

 Click 'Audits' in the top nav
 → Page renders for the user's role



4.2 + Request Audit (INITIATED)

What it does	Buyer creates an audit request with scope + supplier + audit type.
Menu / route	/request-audit
Available to roles	buyer · tenant_admin
Backend API	POST /api/audit-requests

CLICK-BY-CLICK

1. **NAVIGATE** Open /request-audit
→ Form renders
2. **CLICK** Pick supplier from dropdown
→ supplierOrgId
3. **TYPE** Fill scope + auditType (cGMP / ISO 9001 / GDP / ...)
→ required
4. **CLICK** Click 'Submit request'
→ Phase=INITIATED · global HAWK-XXX id assigned

4.3 Supplier decision (INITIATED → PREP / rejected)

What it does	Supplier accepts the audit, declines, or proposes alternate dates.
Menu / route	Audit detail · 'Accept Audit' / 'Propose Dates' / 'Decline' buttons
Available to roles	supplier
Backend API	POST /api/audit-requests/:id/supplier-decision

CLICK-BY-CLICK

1. **CLICK** Click 'Accept Audit'
→ Phase advances to PREP

4.4 Phase advance (auditor / buyer)

What it does	Generic phase advance endpoint with override flag for admins.
Menu / route	Audit detail · 'Advance to next phase' button
Available to roles	auditor · buyer · admin
Backend API	POST /api/audits/:id/phases/transition

CLICK-BY-CLICK

1. **CLICK** Click 'Advance to next phase'
→ Phase moves N → N+1; if artifacts incomplete + not admin → 400 BLOCKED

4.5 AI · Audit-Prep questionnaire (PREP phase)

What it does	Risk-weighted questionnaire pulled from past findings + openFDA + EMA + WHO PQ.
Menu / route	PREP phase workspace · 'Generate questionnaire' button
Available to roles	auditor · admin
Backend API	POST /api/ai/audit-agents/prepare-questionnaire
AI assistance	AI Public-data fusion + Free Gemini for narrative

CLICK-BY-CLICK

1. **CLICK** Click 'Generate questionnaire'
→ After ~3-5s: 6 sections with risk-weighted questions + signals from public data

4.6 AI · Auditor Coach (EXECUTION phase, Wave 3)

What it does	Real-time observation severity guidance per ISO 19011.
Menu / route	Auditor console · side panel during execution

Available to roles	auditor
Backend API	POST /api/ai/auditor-coach/suggest
AI assistance	AI Free Gemini

CLICK-BY-CLICK

1. **TYPE** Auditor types observation in the live console
 - Coach panel suggests severity (Critical / Major / Minor / Observation) + cited clauses

4.7 Deficiency validation (FINDINGS → CAPA)

What it does	Supplier accepts / partially-accepts / disputes findings before CAPA opens.
Menu / route	Audit detail · Findings tab · 'Validate Findings' button (supplier)
Available to roles	supplier
Backend API	POST /api/audit-requests/:id/deficiency-validation

CLICK-BY-CLICK

1. **CLICK** Click 'Validate Findings'
 - Drawer with each finding + accept/dispute toggles
2. **CLICK** Pick deficiencyValidation (ACCEPTED / PARTIALLY_ACCEPTED / DISPUTED)
 - required
3. **TYPE** If DISPUTED, fill disputeReason + disputeltems[]
4. **CLICK** Submit
 - Phase advances to CAPA

4.8 AI · Audit-Report Assembler (FINDINGS / CLOSURE phase)

What it does	Generate findings log PDF + CAPA deadlines per GMP classification + SHA-256 integrity hash.
Menu / route	Findings tab · 'Assemble report' button
Available to roles	auditor
Backend API	POST /api/ai/audit-agents/assemble-report
AI assistance	AI Free Gemini for narrative; SHA-256 for hash

CLICK-BY-CLICK

1. **CLICK** Click 'Assemble report'
 - PDF + JSON returned with SHA-256 contentHash

4.9 Closure certificate (CLOSURE phase)

What it does	Generate + sign 4-role closure certificate (buyer / supplier / auditor / witness).
Menu / route	Closure tab · 'Generate closure cert' + 'Sign' buttons
Available to roles	buyer · supplier · auditor · witness
Backend API	POST /api/audits/:id/artifacts/:certId/sign (per signer)

CLICK-BY-CLICK

1. **CLICK** Click 'Generate closure cert' (buyer)
 - Cert artifact created with 4 pending signatures

2. **CLICK** Each signer clicks 'Sign'
 - Their signature row populated; e-sig captured
3. **WAIT** After 4th signature
 - Phase auto-advances to SURVEILLANCE

5. Full lifecycle walkthrough — one transaction, end-to-end

One audit walked across all 8 phases.

#	Persona	State transition	Action	API	Observed	Result
1	Priya Buyer	— → INITIATED	+ Request Audit · pick supplier · scope · auditType=cGMP	POST /api/audit-requests	Audit request created with HAWK-XXX id · phaseState.currentPhase=INITIATED	PASS

Expected DB state after this step:

```
audit-requests-master { _id, internalRequestId: 'HAWK000000NNN', supplierId, scope, auditType, phaseState: { currentPhase: 'INITIATED' } }
```

state-screens/audit-list-buyer.png

2	Michael (supplier) supplier	INITIATED → PREP	POST supplier-decision · ACCEPTED	POST /api/audit-requests/:id/supplier-decision	Phase=PREP; supplierDecision=ACCEPTED	PASS
3	Maria auditor	PREP → PLANNING	Generate questionnaire (AI) · attach DRL + SMF artifacts · Advance phase	POST /api/ai/audit-agents/prepare-questionnaire · POST /api/audits/:id/phases/transition	Phase=PLANNING; AI returned 6 sections + 4 public signals	PASS
4	Maria auditor	PLANNING → EXECUTION	Build agenda · scope · COI declaration · Advance phase	POST /api/audits/:id/phases/transition	Phase=EXECUTION	PASS

#	Persona	State transition	Action	API	Observed	Result
5	Maria auditor	EXECUTION → FINDINGS	Conduct on-site / remote audit · log observations · use AuditorCoach for severity guidance · close meeting · Advance phase	POST /api/audits/:id/phases/transition	Phase=FINDINGS; observations logged with severity tags	PASS
6	Maria auditor	FINDINGS → FINDINGS (with PDR)	Click 'Assemble report' → AI assembles findings PDF with SHA-256 hash	POST /api/ai/audit-agents/assemble-report	Report PDF ref + contentHash returned	PASS
7	Michael (supplier) supplier	FINDINGS → CAPA	POST deficiency-validation · ACCEPTED	POST /api/audit-requests/:id/deficiency-validation	Phase=CAPA · supplier accepted findings; CAPAs auto-promoted via /candidates/bulk-from-audit	PASS
8	Maria auditor	CAPA → CLOSURE	(After CAPAs closed) Advance phase	POST /api/audits/:id/phases/transition	Phase=CLOSURE	PASS
9	Buyer + supplier + auditor + witness 4 roles	CLOSURE → SURVEILLANCE	Generate closure cert · 4 signatures captured (buyer, supplier, auditor, witness)	POST /api/audits/:id/artifacts/:certId/sign (x4)	Closure cert signed by all 4 roles · phase auto-advances to SURVEILLANCE	PASS

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Audit-Prep Agent (audit-agents)	PREP	POST /api/ai/audit-agents/prepare-questionnaire	PREP workspace · 'Generate questionnaire'	6-section risk-weighted questionnaire pulled from past findings + public FDA / EMA / WHO PQ signals	Public data + Free Gemini
Supplier-Intel Agent	PREP PLANNING	POST /api/ai/audit-agents/supplier-intel	Supplier register drawer	Public FDA warning letters + 483s + recalls + verdict	Public-data fusion
Auditor Coach Panel (Wave 3)	EXECUTION	POST /api/ai/auditor-coach/suggest	Auditor console side panel	Real-time observation severity guidance (Critical/Major/Minor/Observation) per ISO 19011	Free Gemini
Audit-Report Assembler (audit-agents)	FINDINGS CLOSURE	POST /api/ai/audit-agents/assemble-report	Findings tab · 'Assemble report'	Findings log PDF + CAPA deadlines per GMP class + SHA-256 integrity hash	Free Gemini + SHA-256

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
INITIATED	ISO 19011 §6.2	Intimation letter + scope + audit type	audit-requests-master + audit-artifacts
PREP	ISO 19011 §6.3	DRL + Site Master File + pre-audit questionnaire	audit-artifacts + preAuditQuestionnaire
PLANNING	ISO 19011 §6.4.1 EU GMP Chapter 9	Scope + Agenda + COI declaration	audit-artifacts
EXECUTION	ISO 19011 §6.4.5	Opening meeting minutes + observations log + GMP checklist	audit-events + audit-notes
FINDINGS	ISO 19011 §6.4.8	Findings log with severity per ISO 19011 + Preliminary Deficiency Report (PDR) + closing meeting minutes	audit-report + EvidenceFinding
CAPA	21 CFR 820.100 ICH Q10 §3.2.2	Linked CAPAs + supplier evidence	capa-v2 (linked via sourceType=AUDIT_FINDING)
CLOSURE	ISO 19011 §6.5.4 21 CFR Part 11 §11.50	Closure certificate signed by 4 roles (buyer/supplier/auditor/witness)	audit-artifacts (cert) + electronic-signatures
SURVEILLANCE	ISO 19011 §6.6	Follow-up audit scheduling per surveillance cadence	audit-schedule

8. Test results matrix

Suite	Scope	Result	Evidence
novex-walkthrough.spec.ts	/buyer/audits + /auditor/audits register render	PASS	07-priya.png + 11-maria.png
Wave audit-agents smoke	POST /api/ai/audit-agents/prepare-questionnaire · supplier-intel · assemble-report	PASS	audit-agents smoke 9/10 PASS
(deferred) eqms-lifecycle.spec.ts · audit-8-phase	Full 8-phase walkthrough requires buyer+supplier+auditor handoffs	SKIP	Not yet automated end-to-end

9. Known gaps + roadmap

- Full 8-phase E2E lifecycle test** — Need 3-persona orchestration spec (buyer + supplier + auditor) walking each phase + uploading the right artefacts. **HIGH**
- Auto-generate CAPA candidates from major+ findings on FINDINGS phase** — Today buyer must click 'Generate CAPAs from findings'. Auto-fire would close the loop. **MEDIUM**
- Surveillance scheduler** — Cadence-based reminder for follow-up audit per supplier risk band. **MEDIUM**
- Auditor Coach UI panel** — Backend endpoint exists; wire into auditor console. **MEDIUM**